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RAMAIAH
Institute of Technology
(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – JULY/AUGUST 2024

Program	: **B.E. – Common to CSE / ISE / CSE(CY) / AI & DS / BT/ AI & ML/ CSE (AI & ML)**	**Semester**	:	**II**
Course Name	: **Engineering Chemistry**	**Max. Marks**	:	**100**
Course Code	: **CYC22**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:
- Answer one full question from each unit.
- Draw neat labeled diagram, wherever necessary.

UNIT - I

1. a) What is the difference between Galvanic and Electrolytic cell? Explain the origin of single electrode potential. CO1 (08)
   b) Derive the Nernst equation for the following reaction and explain the terms involved: CO1 (06)
   $$Fe^{3+} + e^{-} \Leftrightarrow Fe^{2+}$$
   c) List the Key components of Ni-MH_2 battery and write its anodic and cathodic reactions during discharge. CO1 (06)

2. a) What is the significance of the Electrode potential values? Explain the construction and working of Calomel electrode with neat labelled diagram. CO1 (08)
   b) A silver electrode dipped in a 1.00 M solution of $AgNO_3$ was connected by a means of salt bridge to a copper electrode dipped in effluent sample. The Cell potential measured was 0.52 V. What is the concentration of Cu^{2+} ion in the water sample? Given E^oAg/Ag^+ =0.8V; E^oCu^{2+}/Cu=0.34 V. CO1 (06)
   c) Explain the following characteristics of battery: CO1 (06)
   (i) Capacity (ii) Shelf life (iii) Cycle life.

UNIT - II

3. a) Discuss the rusting of iron with the help of electrochemical theory of wet corrosion. CO2 (08)
   b) What are corrosion inhibitors? Discuss corrosion inhibition by cathodic inhibitors with relevant reactions involved. CO2 (06)
   c) Assess the following factors which influence the rate of corrosion: CO2 (06)
   (i) Relative size of anodic and cathodic area (ii) pH.

4. a) Describe the anodizing of aluminum with suitable reactions and mention it's any two applications. CO2 (08)
   b) What is cathodic protection? Explain the sacrificial anodic method to prevent corrosion. CO2 (06)
   c) Give reasons: CO2 (06)
   (i) Galvanized containers are not used for storage of food products, but tin coated containers are used.
   (ii) Iron corrodes under drops of salt solution.

UNIT - III

5. a) What is Glass transition temperature (T_g)? Explain how flexibility, intermolecular forces and molecular weight affect the T_g of a polymer? CO3 (08)
   b) What are Liquid crystals? Give its classification with relevant example for each. CO3 (06)
   c) How the molecular arrangement in solids, liquids and liquid crystals vary? Explain the significance of director in liquid crystals. CO3 (06)

6.	a)	What are conducting polymers? Elucidate the stepwise mechanism of conduction in polyacetylene.	CO3	(08)
	b)	How (i) Teflon (ii) PMMA are synthesized? Explain any two properties and applications for each.	CO3	(06)
	c)	Explain the liquid crystalline behavior in PAA with neat diagram.	CO3	(06)

UNIT- IV

7.	a)	State Beer-Lambert's Law. Elaborate on the principle and procedure involved in estimating copper colorimetrically.	CO4	(08)
	b)	Explain the synthesis of nanomaterials by solution combustion method.	CO4	(06)
	c)	Mention characterization technique and applications of nanomaterials.	CO4	(06)
8.	a)	Elaborate on the principle and procedure involved in estimating Iron using potentiometric sensor.	CO4	(08)
	b)	Explain hydrothermal method of synthesis of nanomaterials.	CO4	(06)
	c)	Outline the determination of pH of solution using pH meter.	CO4	(06)

UNIT - V

9.	a)	What are PV cells? Outline the construction and working of PV cells. List any two advantages of PV cells.	CO5	(08)
	b)	Explain the construction and working of polymer electrolyte membrane fuel cells.	CO5	(06)
	c)	List and explain the effective methods of disposal of e-waste.	CO5	(06)
10.	a)	How fuel cells are different from batteries? With neat, labelled diagram explain the construction and working of methanol-oxygen fuel cells.	CO5	(08)
	b)	List toxic metals present in e-waste and their effect on environment and human health.	CO5	(06)
	c)	Mention the sources and chemical composition of e-waste.	CO5	(06)

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